

FUNCTIONS OF LIVER

Headings

▼ FUNCTIONS OF LIVER

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- Liver is the largest gland in the human body.

1. METABOLIC FUNCTIONS

- Major site for metabolic reactions including the metabolism of:
 - a. **Carbohydrates**
 - b. **Proteins**
 - c. **Fats (Lipids)**
 - d. **Vitamins**

2. STORAGE FUNCTION

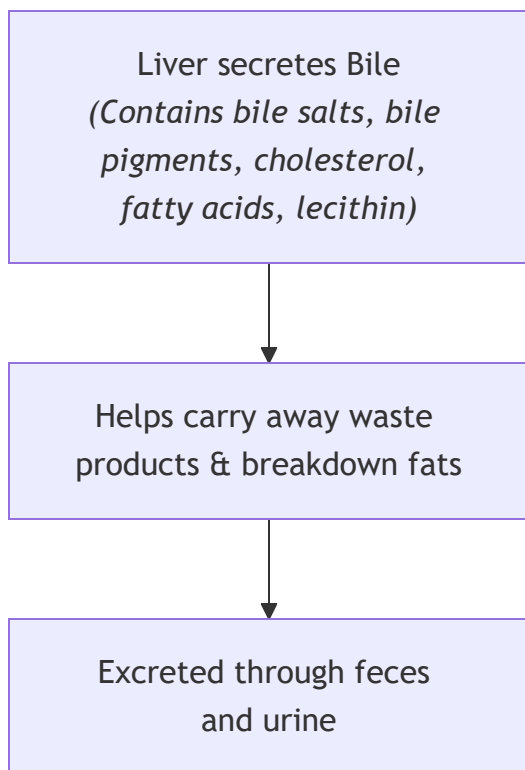
- Stores essential nutrients and vitamins:
 - a. **Glycogen**
 - b. **Amino acids (AA)**
 - c. **Iron**
 - d. **Folic acid**

- e. **Vitamins:** Vitamin A, Vitamin B₁₂, and Vitamin D

3. SYNTHETIC FUNCTION

- a. **Glucose Production** : Via **gluconeogenesis**.
- b. **Plasma Proteins** : Synthesizes all plasma proteins & other proteins (*except Immunoglobulins*):
 - Clotting factors
 - Complement factors
 - Hormone-binding proteins
- c. **Other Substances** : Synthesizes **steroids, heparin, and somatomedin (IGF-1)**.

4. SECRETION OF BILE



5. EXCRETORY FUNCTION

- Excretes various end products and foreign substances:
 - a. **Cholesterol**
 - b. **Bile pigments** (Bilirubin & Biliverdin)
 - c. **Bacteria**
 - d. **Heavy metals** (Pb, As, Bi)
 - e. **Toxins**
 - f. **Viruses** (e.g., *Yellow fever virus*)

6. HEAT PRODUCTION

- An **enormous amount of heat** is produced due to high metabolic activity (*site of maximum heat production in the body*).

7. HEMOPOIETIC FUNCTION

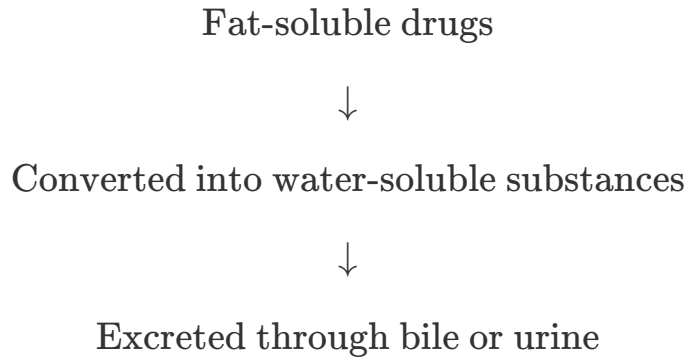
- a. **In Fetus** : Produces blood cells during the **hepatic stage of fetal life**.
- b. **Storage** : Stores **Vitamin B₁₂** (necessary for erythropoiesis) and **Iron** (necessary for hemoglobin synthesis).
- c. **Thrombopoietin Production** : Produces **thrombopoietin**, which promotes thrombocyte (platelet) production.

8. HEMOLYTIC FUNCTION

- **Destruction of Senile RBCs** : Senile RBCs (after their lifespan of ≈ 120 days) are destroyed by **reticuloendothelial cells (Kupffer cells)** of the liver.

9. INACTIVATION OF HORMONES AND DRUGS

- a. **Catabolism of Hormones** : Catabolizes Growth Hormone, Parathormone, Cortisol, Insulin, Glucagon, and Estrogen.
- b. **Inactivation of Drugs** :



10. DEFENSIVE AND DETOXIFICATION FUNCTIONS

- **Defensive Function** :
- **Kupffer cells** (reticuloendothelial cells) defend against infections via phagocytosis.
- **Detoxification of Foreign Bodies** :
- a. Foreign bodies (bacteria or antigens) are swallowed and digested by reticuloendothelial cells (**phagocytosis**).
- b. Reticuloendothelial cells produce cytokines like **interleukins** and **tumor necrosis factor (TNF)** to activate the immune system.
- c. Liver cells remove toxic properties of harmful agents (**detoxification**) via:
 - Total destruction by means of **metabolic degradation**.
 - Conversion of toxic substances into non-toxic materials by means of **conjugation**.